

ECO351: Homework 3

Question 1. Truncation and Censoring.

- (a) Substituting $\mu_i = \exp(x'_i\beta)$ to parametrize the pdf, we get the PDF conditional on x_i :

$$f(y_i|x_i) = \mu_i^{-1} \exp(-y_i/\mu_i)$$

$$\boxed{f(y_i|x_i) = \exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta))} \quad y_i \geq 0$$

Similarly, the CDF can be written by substituting the same value:

$$F(y_i|x_i) = 1 - \exp(-y_i/\mu_i)$$

$$\boxed{F(y_i|x_i) = 1 - \exp(-y_i \exp(-x'_i\beta))}$$

- (b) The probability that a worker is employed for more than 36 months is given by,

$$\begin{aligned} Pr(y_i > 36|x_i) &= 1 - Pr(y_i \leq 36|x_i) \\ &= 1 - F(36|x_i) \\ &= 1 - [1 - \exp(-36 \exp(-x'_i\beta))] \end{aligned}$$

$$\boxed{Pr(y_i > 36|x_i) = \exp(-36 \exp(-x'_i\beta))}$$

- (c) The sample is **truncated** since we do not observe x_i and y_i for $y_i \leq 36$. This implies that we cannot utilise the information for individuals who have been employed for less than or equal to 36 months. It can be represented as:

$$\begin{aligned} z_i &= x'_i\beta + \varepsilon, \varepsilon \sim \text{Exp}(\exp(x'_i\beta)) \\ y_i &= \begin{cases} z_i & \text{if } z_i > 36 \\ \text{unobserved} & \text{otherwise} \end{cases} \end{aligned}$$

The **individual likelihood** is:

$$\begin{aligned} l(\beta) &= f(y_i|y_i > 36, x_i) \\ &= \frac{f(y_i|x_i)}{Pr(y_i > 36|x_i)} \\ &= \frac{f(y_i|x_i)}{1 - F(36|x_i)} \\ &= \frac{\exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta))}{\exp(-36 \exp(-x'_i\beta))} \end{aligned}$$

$$\boxed{l(\beta) = \exp(-x'_i\beta - (y_i - 36) \exp(-x'_i\beta))}$$

For the **individual log-likelihood function**, we take log on both sides.

$$\boxed{\ln l(\beta) = -x'_i\beta - (y_i - 36) \exp(-x'_i\beta)}$$

The **sample log-likelihood function** can be derived by taking the logarithm of the sample likelihood function. Here, the number of individuals is $\tilde{n}$. Let the sample likelihood function be $L(\beta)$

$$L(\beta) = \prod_{i=1}^{\tilde{n}} \exp(-x'_i\beta - (y_i - 36)\exp(-x'_i\beta))$$

$$\ln L(\beta) = \ln\left(\prod_{i=1}^{\tilde{n}} \exp(-x'_i\beta - (y_i - 36)\exp(-x'_i\beta))\right)$$

$$\boxed{\ln L(\beta) = \sum_{i=1}^{\tilde{n}} \left[-x'_i\beta - (y_i - 36)\exp(-x'_i\beta) \right]}$$

(d) This model can be called **censored** since we do observe the x_i s for all workers in the company instead of just those who have been employed for more than 36 months.

The **individual contributions** are:

- For an observed (uncensored) worker i with observed $y_i > 36$:

$$l_i(\beta) = f(y_i|x_i)$$

$$= \exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta))$$

- For a censored worker i for whom we only know $y_i \leq 36$:

$$l_i(\beta) = \Pr(y_i \leq 36|x_i)$$

$$= F(36|x_i)$$

$$= 1 - \exp(-36 \exp(-x'_i\beta))$$

The **individual log-likelihoods** are:

- For an observed (uncensored) worker i with observed $y_i > 36$:

$$\ln l_i(\beta) = \ln(\exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta)))$$

$$= -x'_i\beta - y_i e^{-x'_i\beta}$$

- For a censored worker i for whom we only know $y_i \leq 36$:

$$\ln l_i(\beta) = \ln(1 - \exp(-36 \exp(-x'_i\beta)))$$

The sample likelihood is given as:

$$L(\beta) = \prod_{i:y_i > 36} f(y_i|x_i) \times \prod_{j:y_j \leq 36} F(36|x_j)$$

The **sample log-likelihood** is calculated as below:

$$\ln L(\beta) = \ln\left(\prod_{i:y_i > 36} f(y_i|x_i) \times \prod_{j:y_j \leq 36} F(36|x_j)\right)$$

$$= \sum_{i:y_i > 36} [-x'_i\beta - y_i e^{-x'_i\beta}] + \sum_{j:y_j \leq 36} \ln(1 - \exp(-36 \exp(-x'_j\beta)))$$

Question 2. Tobit Type I Model.

a) **Number and percentage of observations.** The number and percentage of observations in each program category are summarized below:

Program Type	Number of Observations	Percentage (%)
Academic	45	22.50
General	105	52.50
Vocational	50	25.00

TABLE 1. Distribution of program types among 200 observations.

b) **Descriptive Summary.** The descriptive summary for the three variables *read*, *math*, and *apt* is presented below. The table reports the mean, median, standard deviation, minimum, and maximum values for each variable.

Variable	Mean	Median	Std. Dev.	Min	Max
Read	52.23	50	10.25	28	76
Math	52.65	52	9.37	33	75
Apt	640.03	633	99.22	352	800

TABLE 2. Descriptive summary statistics for *read*, *math*, and *apt*.

c) **Histogram of apt.** The histogram shows the distribution of **aptitude scores** across the three program type.

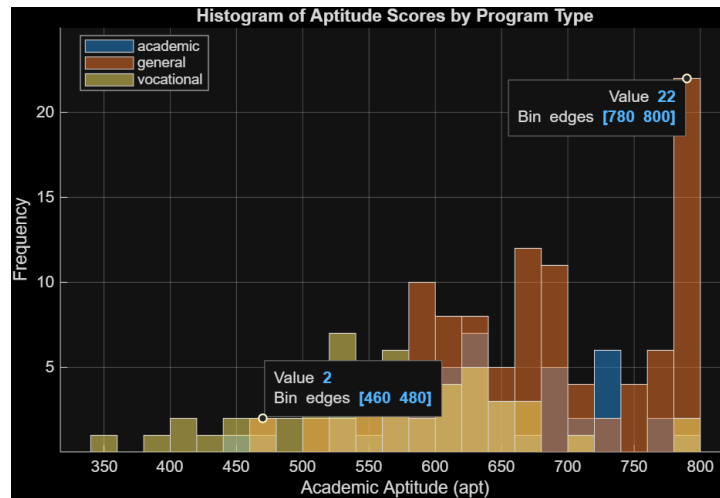


FIGURE 1. Histogram of Academic Aptitude (apt) Scores.

The histogram reveals distinct yet overlapping distributions of **Academic Aptitude (apt)** scores across the three program types: academic, general, and vocational.

- The **academic program** (blue) scores are predominantly concentrated at the **higher end**, with a clear mode between 750 and 800, suggesting students in this program tend to have the highest aptitude.
- The **vocational program** (olive/gold) scores are generally **lower**, with a greater frequency in the middle-to-lower range, spanning approximately 450 to 650. Its distribution appears somewhat left-skewed relative to the other two.

- The **general program** (brown) exhibits the **widest spread**, covering the entire range of scores (~ 400 to 800). It has a high frequency across the distribution but also has the largest absolute frequency in the 750 to 800 bin, indicating a large number of high-aptitude students are in this category.

Overall, there is considerable **overlap** in the middle aptitude range (~ 550 to 700) for all three programs, but the academic and vocational programs show a clear separation in their central tendencies towards the upper and lower halves, respectively.

d) **Correlation table and Scatter plots.** The **correlation matrix** for the continuous variables *read*, *math*, and *apt* is presented below.

Variable	Read	Math	Apt
Read	1.000	0.662	0.645
Math	0.662	1.000	0.733
Apt	0.645	0.733	1.000

TABLE 3. Correlation matrix for Read, Math, and Aptitude scores.

The correlation coefficients indicate a strong positive relationship among all three variables. The **scatter plots** between the pairs of continuous variables are shown below.

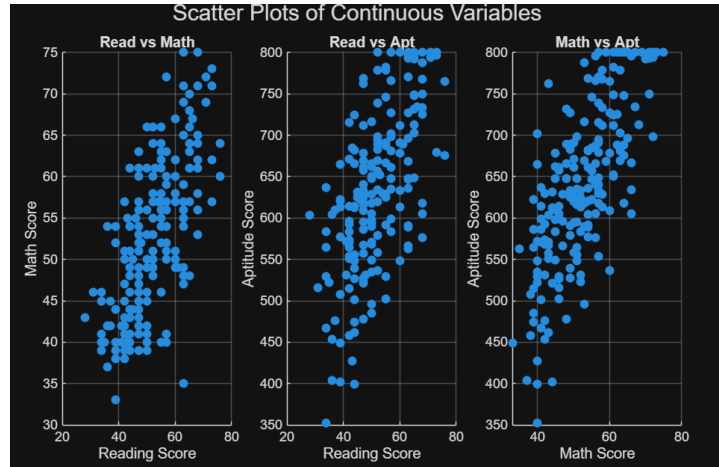


FIGURE 2. Scatter plots.

Observation: All three scatter plots indicate a positive relationship among the variables. Higher reading and math scores are associated with higher aptitude scores. In the "Read vs Apt" plot and the "Math vs Apt" plot, the data points for Aptitude Score (the y-axis) cluster densely at the very top, around 800. This suggests that 800 is the maximum possible score, and for individuals who performed well enough to score above 800, their recorded score is capped at 800. Therefore, **Aptitude score is censored from above at 800.**

e) **Tobit Type I Model Estimation.** The Tobit Type I model (also known as the Censored Regression Model) is used when the dependent variable is censored at a certain threshold but the underlying latent variable is assumed to follow a normal distribution. In this case, the dependent variable is *aptitude* (*apt*), which represents students' academic aptitude scores. The model assumes that there exists an unobserved latent variable apt_i^* related to the observed variables as:

$$apt_i^* = \beta_0 + \beta_1 read_i + \beta_2 math_i + \beta_3 general_i + \beta_4 vocational_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

and the observed variable follows:

$$apt_i = \begin{cases} apt_i^*, & \text{if } apt_i^* < 800, \\ 800, & \text{if } apt_i^* \geq 800. \end{cases}$$

The model parameters are estimated using Maximum Likelihood Estimation (MLE) via the `fminunc` optimizer in MATLAB. The academic program category serves as the base group.

Variable	Estimate	Std. Error	t-Statistic
Constant	209.57	32.774	6.3943
Read	2.6979	0.6188	4.3600
Math	5.9145	0.7099	8.3314
General	-12.715	12.406	-1.0249
Vocational	-46.144	13.724	-3.3622
log(σ)	4.1847	0.0530	78.901

TABLE 4. Tobit Type I Model Estimates (Right-Censored at 800)

Interpretation:

- The coefficients of both *read* and *math* are **positive and statistically significant** at the 5% level ($|t| > 1.96$), implying that higher reading and mathematics scores are associated with higher latent academic aptitude. Specifically, holding other factors constant, a one-unit increase in the reading score increases the latent aptitude by approximately 2.70 units, while a one-unit increase in the mathematics score increases it by about 5.91 units.
- The dummy variable for the general program is negative but statistically insignificant, indicating that students in the general program do not differ significantly in aptitude from those in the academic program.
- In contrast, the coefficient for the vocational program is negative and statistically significant, suggesting that, on average, students enrolled in the vocational program have lower aptitude scores (approximately 46 units less) than those in the academic program, after controlling for reading and mathematics performance.

Overall, the Tobit Type I results suggest that both reading and math proficiency are strong predictors of students' academic aptitude, while program type plays a significant role in explaining variation across groups.

f) Covariate Effects. The Tobit Type I latent variable specification is

$$apt_i^* = \beta_0 + \beta_1 read_i + \beta_2 math_i + \beta_3 general_i + \beta_4 vocational_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

and the observed outcome is right-censored at $c = 800$:

$$apt_i = \begin{cases} apt_i^*, & \text{if } apt_i^* < 800, \\ 800, & \text{if } apt_i^* \geq 800. \end{cases}$$

Define $z_i = \frac{c - x_i' \beta}{\sigma} = \frac{800 - x_i' \beta}{\sigma}$. The conditional expectation of the *observed* aptitude is

$$E[apt_i | x_i] = \Phi(z_i) x_i' \beta - \sigma \varphi(z_i) + c(1 - \Phi(z_i)),$$

where $\Phi(\cdot)$ and $\varphi(\cdot)$ are the standard normal CDF and PDF. Intuitively, the first two terms give the contribution from uncensored (latent) values and the third term accounts for the mass of observations censored at c .

The marginal effect of a continuous covariate x_{ij} on the expected *observed* aptitude equals

$$\frac{\partial E[apt_i | x_i]}{\partial x_{ij}} = \beta_j \Phi(z_i).$$

Thus the effect on the observed (censored) outcome equals the Tobit coefficient β_j scaled by the probability of being *uncensored* (i.e. $P(\text{apt}_i^* < 800 \mid x_i) = \Phi(z_i)$). The Average Marginal Effect (AME) is

$$AME_j = \frac{1}{N} \sum_{i=1}^N \beta_j \Phi(z_i).$$

For a binary covariate (e.g. the vocational dummy) it is conventional to report the average discrete change:

$$AME_{\text{voc}} = \frac{1}{N} \sum_{i=1}^N \left[E(\text{apt}_i \mid \text{prog} = \text{voc}, x_i) - E(\text{apt}_i \mid \text{prog} = \text{academic}, x_i) \right].$$

Using the estimates from part (e) ($\hat{\beta}_{\text{Read}} = 2.697$, $\hat{\beta}_{\text{Math}} = 5.914$, $\hat{\beta}_{\text{Voc}} = -46.144$, $\hat{\sigma} = \exp(4.205) \approx 66.9$):

- **Reading:** The average marginal effect on the *observed* aptitude for observation i is approximately 2.501 which means the expected change in *apt* for a one unit increase in *read* is 2.501.
- **Mathematics:** Similarly, the average marginal effect due to mathematics on observed aptitude is 5.482.
- **Vocational program (discrete change):** The average effect of switching from academic to vocational on the *observed* expected aptitude for an observation i is -43.838.

(g) **Scatter Plots of Residuals and Fitted Values.**

- The scatter plot shows a distinct cluster of points forming a negatively sloped boundary at the upper fitted values (around 700–800).
- These red points correspond to censored observations, where the latent aptitude would have been higher than 800 but is censored at the limit.
- The uncensored points (blue) appear more symmetrically scattered around zero for lower fitted values, indicating a roughly homoskedastic residual pattern for the uncensored portion.
- The sharp cutoff among red points reflects the censoring mechanism, a classic signature of an upper-censored Tobit model.
- Therefore, the residuals show a clear truncation boundary for censored observations at 800, confirming that censoring is correctly modeled. Uncensored residuals appear randomly distributed, indicating no major specification issues.

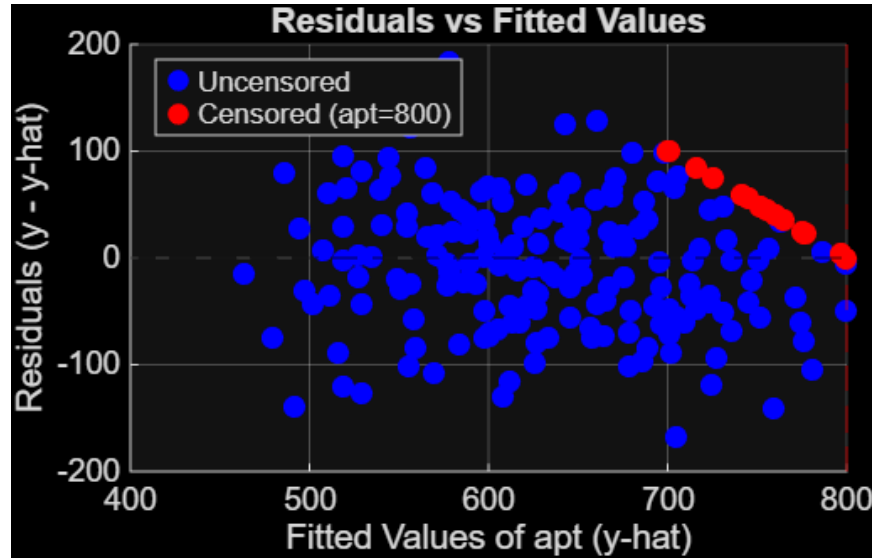


FIGURE 3. Residuals vs. Fitted Values of *apt*

- The observed–fitted plot shows that most points lie close to the 45° line for uncensored observations, suggesting a good overall model fit.
- However, the censored cases (in red) all pile up at the top ($y = 800$), deviating from the 45° line.

- This concentration at the upper limit again reflects right-censoring, as the true latent aptitudes for these individuals exceed the observable maximum.
- Therefore, the fitted values align closely with observed aptitude for uncensored cases, while the flat cluster at the top indicates censoring at 800. This pattern is consistent with a correctly specified Tobit model.

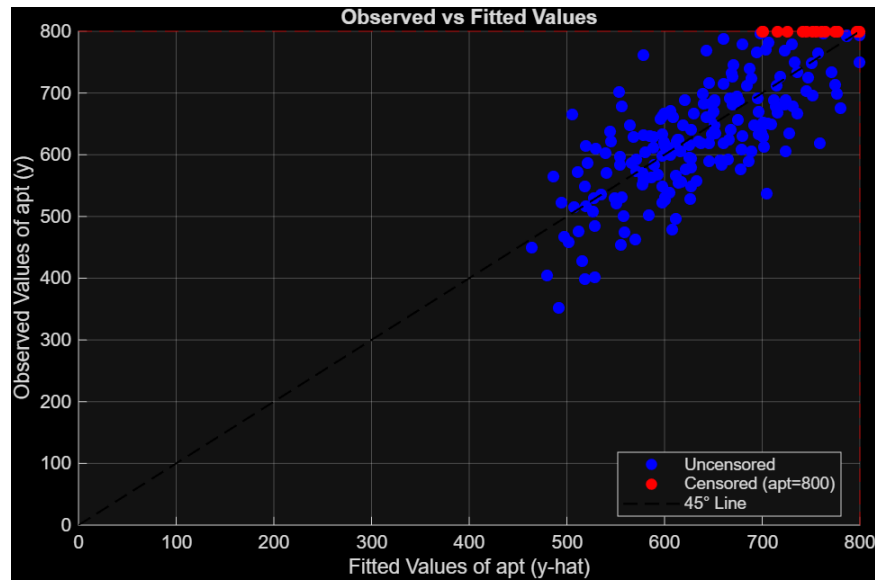


FIGURE 4. Observed *apt* vs. Fitted Values of *apt*

Question 3.

a) OLS. The regression model we are using with the the independent variables **educ**, **exper**, and **hhours** which are her education, her experience, and her husband's hours of work to explain the woman's hours of work. The model is represented as follows:

$$(1) \quad \text{hours} = \beta_1 + \beta_2 \text{educ} + \beta_3 \text{exper} + \beta_4 \text{hhours} + \varepsilon_i$$

The estimates for the above model using OLS are:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Intercept	β_1	-20.6750	190.7690	-0.108
Educ	β_2	31.1346	12.7894	2.434
Exper	β_3	42.8949	3.6208	11.847
Hhours	β_4	-0.0341	0.0491	-0.694

These OLS estimates are not consistent specifically because we include the data of Women who are not part of the labor force (i.e $lfp = 0$) which introduces a very large number of zeroes. These unusually large number of zero-working hours even when other variables are consistent with that of high working hours(for example education) impacts the coefficient estimate incorrectly. For example: Two individuals with the same education level but with one being part of the labor force and the other not, in this case the coefficient is impacted negatively and hence it will be inconsistent. Therefore, the inflated number of zeroes makes the coefficient estimates inconsistent.

b) OLS with selection. While estimating this model using OLS we only consider the women whose working hours > 0 . The estimates for the above model using OLS are:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Intercept	β_1	1143.9	245.24	4.6645
Educ	β_2	-21.021	15.779	-1.3322
Exper	β_3	28.883	4.4735	6.4565
Hhours	β_4	0.021734	0.0621033	0.34997

The OLS estimates in this case are also not consistent as there is a sampling selection bias which leads to the error term being correlated with the selection rule of $hours > 0$.

c) Estimating Probit model. The following probit model is used for estimating women's decision to be in the labour force i.e $LFP = 1$:

This is the selection model, the participation latent variable(z_{1i}) equation can be written as:

$$(2) \quad z_{1i} = \gamma_1 + \gamma_2 \text{exper} + \gamma_3 \text{kidsl6} + \gamma_4 \text{kids618} + \gamma_5 \text{MTR} + \gamma_6 \text{largecity} + \varepsilon_{1i}$$

$$(3) \quad LFP_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$(4) \quad \Pr(LFP = 1) = \Pr(z_i > 0) = \Phi(\gamma_1 + \gamma_2 \text{exper} + \gamma_3 \text{kidsl6} + \gamma_4 \text{kids618} + \gamma_5 \text{MTR} + \gamma_6 \text{largecity})$$

The following are the coefficient estimates, standard errors and t-stats for the above model:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Intercept	γ_1	1.39	0.44472	3.1255
Exper	γ_2	0.064384	0.00707	9.094
kidsl6	γ_3	-0.39382	0.09645	-4.0832
kids618	γ_4	0.14778	0.03977	3.7158
MTR	γ_5	-2.755	0.62034	-4.441
largecity	γ_6	-0.16179	0.10604	-1.5258

As we can see the coefficients of Experience, Kids below 6, Kids between 6 and 18, and MTR have significant t-statistics and hence play an important role in the woman's decision to participate in the labor force. The coefficient of experience is still significantly lower in value than the rest, hence we can also say that the other variable play a more crucial role in explaining the decision.

The variables MTR and kids below 6 have a negative effect on the decision while kids between 6 to 18 has a positive effect on the decision.

d). Using the estimates from the probit model we are to obtain:

$$(5) \quad \tilde{w} = \tilde{\gamma}_1 + \tilde{\gamma}_2 \text{ exper} + \tilde{\gamma}_3 \text{ kidsl6} + \tilde{\gamma}_4 \text{ kids618} + \tilde{\gamma}_5 \text{ MTR} + \tilde{\gamma}_6 \text{ largacity}$$

And using this, we create the inverse Mills ratio

$$(6) \quad \tilde{\lambda} = \frac{\varphi(\tilde{w})}{\Phi(\tilde{w})}$$

$$(7) \quad \text{Sample Mean of } \tilde{\lambda} = 0.716302$$

$$(8) \quad \text{Sample Variance of } \tilde{\lambda} = 0.1173$$

e) Estimating the hours model. The following model is to be estimated

$$(9) \quad \text{hours} = \beta_1 + \beta_2 \text{educ} + \beta_3 \text{exper} + \beta_4 \text{hhours} + \beta_\lambda \tilde{\lambda} + \varepsilon$$

And we use the observations for which hours > 0.

The following are the estimates for the above model to estimate hours:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Intercept	β_1	2094.6	340.27	6.1557
Educ	β_2	-25.807	15.56	-1.6585
Exper	β_3	5.4648	7.3705	0.74144
hhours	β_4	-0.029992	0.062438	-0.48035
Lambda	β_λ	-797.7	201.47	-3.9594

Including $\tilde{\lambda}$ corrects the bias from estimating hours only for women in the labor force as we did in part (b). Since the coefficient of $\tilde{\lambda}$ is statistically significant, there was a sample selection bias which is being corrected by including $\tilde{\lambda}$. Suppose, if it wasn't significant then the OLS estimation from part(b) would have been unbiased and consistent. That is clearly not the case.

The standard errors are still not correct because the error term is heteroscedastic conditional on selection, hence violating homoscedastic OLS assumptions.

f) Using robust Standard Errors. We estimate the above model using heteroscedasticity robust standard errors to obtain the heteroscedastically consistent estimates and standard errors.

$$\begin{aligned} \hat{\Omega}_{n*n} &= \text{diag}[\hat{\varepsilon}_1, \hat{\varepsilon}_2, \dots, \hat{\varepsilon}_n] \\ \hat{\beta}_{FGLS} &= (X' \hat{\Omega}^{-1} X)^{-1} (X' \hat{\Omega}^{-1} y) \\ \hat{V}(\hat{\beta}) &= (X' X)^{-1} (X' \hat{\Omega} X) (X' X)^{-1} \end{aligned}$$

The above estimates are as follows:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Constant	β_1	2094.6	349.04	6.0012
Educ	β_2	-25.807	15.067	-1.7127
Exper	β_3	5.4648	7.306	0.7480
hhours	β_4	-0.029992	0.064558	0.4646
Lambda	β_λ	797.7	198.81	-4.0124

g) Bootstrap Errors. Following are the estimates with the Bootstrap standard errors:

Variable	Coefficients	Estimates	Bootstrap Standard Errors	t-statistic
Intercept	β_1	2094.6	412.54	5.0774
Educ	β_2	-25.807	16.276	-1.5855
Exper	β_3	5.4648	9.0439	0.60425
hhours	β_4	-0.029992	0.060781	-0.49344
Lambda	β_λ	-797.7	215.99	-3.1656

The bootstrap standard errors are larger than the standard errors in part (e) because the naive OLS Standard Errors contain lambda which is generated and hence violates homoscedasticity. The bootstrap SEs are larger than the estimates in (e) and (f) but they are still closer to the values in (f).

The bootstrap SEs affect the t-statistics, and it reduces the significance of variables with marginal significance in (e). For example, the significance of Education falls below 1.96. The inverse-mills ratio is still significant and hence indicates that sample selection bias is present. The bootstrap SE accounts for the estimation of the inverse mills ratio and the heteroskedasticity in the two step procedure.

h) Heckman's Sample Selection Model. Equation (1) for hours is the outcome equation and equation (4) is used as the selection criteria. We use the full information maximum likelihood to compute the estimates which are as follows:

Variable	Coefficients	Estimates	Standard Errors	t-statistic
Outcome Intercept	β_1	1145.4	286.31	4.0007
Educ	β_2	-5.0625	15.77	-0.32103
Outcome -Exper	β_3	22.051	5.7161	3.8578
hhours	β_4	0.046664	0.062431	0.74745
Selection Constant	γ_1	1.6664	0.44587	3.7374
Selection Exper	γ_2	0.0642	0.0071252	9.0102
Kidsl6	γ_3	-0.42455	0.095034	-4.4674
Kids618	γ_4	0.13877	0.039936	3.4747
MTR	γ_5	-3.1008	0.62052	-4.997
Largecity	γ_6	-0.19052	0.10383	-1.835
Log of s.d	$\ln \sigma$	6.6417	0.046454	142.97
Correlation	$atanh(\rho)$	-0.42026	0.17395	-2.416

The values of σ and ρ are as follows:

$$\hat{\sigma} = 766.40468$$

$$\hat{\rho} = -0.3972$$

The Heckman estimates indicate the same as the estimates in (e) in terms of the direction of the effect of the given variables. Some coefficients like that of experience are larger in value and significance due to the efficiency of the FIML, which maximizes the joint likelihood.

Some of the coefficient estimates in (e) are larger than those in FIML due to the outcome variance changing the coefficient scale.

The bias term which is λ in (e) and ρ in (h) are significant in both and negative as well which confirms the selection bias. Overall the two-step model provides consistent but less efficient estimates and the FIML produces a more reliable specification.

Question 4.

a) Fraction receiving job training : To determine the proportion of individuals who participated in job training, we used the indicator variable **train**, which takes the value 1 if a man received job training between 1975–1977, and 0 otherwise. The variable **train** is binary, so its mean represents the proportion of workers who underwent training:

$$\text{Fraction Trained} = \frac{\sum_{i=1}^N \text{train}_i}{N}$$

From the computation, we obtain:

$$\text{Number trained} = 185, \quad \text{Total sample size} = 445$$

$$\Rightarrow \text{Fraction trained} = \frac{185}{445} \approx 0.416$$

$\Rightarrow$ Approximately 41.6% of the men in the sample received job training during the 1975–1977 period. This fraction will serve as the treatment proportion in subsequent analysis of training effects on 1978 earnings

(b) Difference on receiving job training: The variable **re78** represents real earnings in 1978, measured in thousands of 1982 dollars. Comparing the average earnings of individuals who received training (**train** = 1) with those who did not (**train** = 0). The computed sample means are:

$$\bar{y}_{\text{trained}} = 6.349, \quad \bar{y}_{\text{untrained}} = 4.555, \quad \Rightarrow \Delta = 6.349 - 4.555 = 1.794$$

$\Rightarrow$ On average, men who received job training earned approximately \$1,794 more in 1978 (in 1982 dollars) than those who did not receive training. Considering the relatively low-income nature of the sample, this difference is economically meaningful. It indicates that participation in job training programs was associated with noticeably higher post-training earnings, suggesting a potentially positive treatment effect.

(c) Effect of Job Training in Unemployment : The variable **unem78** is a binary indicator taking the value 1 if an individual was unemployed in 1978, and 0 otherwise. To compute the unemployment fractions, we separate the sample based on the job training indicator **train** and take the mean of **unem78** within each group.

$$\text{Fraction unemployed (trained)} = \frac{\sum_{i:T_i=1} \text{unem78}_i}{N_{T=1}} = 0.243$$

$$\text{Fraction unemployed (untrained)} = \frac{\sum_{i:T_i=0} \text{unem78}_i}{N_{T=0}} = 0.354$$

$$\text{Difference} = 0.243 - 0.354 = -0.111$$

Hence, about 24.3% of trained men were unemployed in 1978, compared to 35.4% among those who did not receive training. The difference of -0.111 indicates that job training is associated with an 11.1 percentage point reduction in unemployment, suggesting a potentially meaningful economic effect.

	Trained	Untrained	Difference
Earnings	6.349	4.555	+1.794
Unemployment	0.243	0.354	−0.111

(d) Overall effectiveness of Job Training program. From parts (b) and (c) , these findings suggest that the job training program may have had a **positive economic impact**, both in terms of **increasing earnings** and **reducing unemployment**. However, since this analysis is purely based on mean comparisons, it does not account for potential selection bias or other confounding factors that might influence both training participation and outcomes.

To strengthen our conclusions, we could control for observable factors like education and prior earnings using regression analysis, or apply causal inference methods such as instrumental variables to better isolate the true treatment effect.

(e) Likelihood function of Treatment Effect model: The outcome equation of Treatment effect model is specified as:

$$y_i = x'_{2i}\beta_2 + \gamma T_i + \varepsilon_{2i}$$

Here, T_i is the treatment indicator, x_{1i} and x_{2i} are vectors of covariates, γ captures the treatment effect, and the error terms $(\varepsilon_{1i}, \varepsilon_{2i})$ may be correlated, introducing potential selection bias.

The likelihood is the probability of observing the data y_i, T_i given the parameters. The joint likelihood for the pair (y_i, T_i) can be expressed as:

$$L_i = \begin{cases} \Phi_2\left(x'_{1i}\beta_1, \frac{y_i - x'_{2i}\beta_2 - \gamma}{\sigma_2}; \rho\right), & \text{if } T_i = 1, \\ \Phi_2\left(-x'_{1i}\beta_1, -\frac{y_i - x'_{2i}\beta_2}{\sigma_2}; \rho\right), & \text{if } T_i = 0, \end{cases}$$

where $\Phi_2(\cdot, \cdot; \rho)$ denotes the cumulative distribution function of the bivariate standard normal distribution with correlation coefficient $\rho = \sigma_{12}/\sigma_2$.

The corresponding log-likelihood function for the full sample is given by:

$$\ln L(\theta) = \sum_{i=1}^N \ln L_i(\theta), \quad \text{where } \theta = (\beta_1, \beta_2, \gamma, \sigma_2, \rho).$$

(f) Estimation of the Treatment Effect Model. The treatment and outcome equations are specified as follows:

$$(10) \quad \begin{aligned} \mathbf{train} = & \beta_{10} + \beta_{11}\text{age}_i + \beta_{12}\text{educ}_i + \beta_{13}\text{black}_i + \beta_{14}\text{hisp}_i + \beta_{15}\text{married}_i \\ & + \beta_{16}\text{nodegree}_i + \beta_{17}\text{re74}_i + \beta_{18}\text{unemp74}_i + \varepsilon_{1i}, \end{aligned}$$

$$\mathbf{train} = \begin{cases} 1 & \text{if } T_i^* > 0, \\ 0 & \text{otherwise.} \end{cases}$$

$$(11) \quad \begin{aligned} \mathbf{re78} = & \beta_{20} + \beta_{21}\text{age}_i + \beta_{22}\text{educ}_i + \beta_{23}\text{black}_i + \beta_{24}\text{hisp}_i + \beta_{25}\text{married}_i \\ & + \beta_{26}\text{re74}_i + \beta_{27}\text{re75}_i + \gamma T_i + \varepsilon_{2i}. \end{aligned}$$

The error terms are assumed to follow a joint normal distribution:

$$\begin{pmatrix} \varepsilon_{1i} \\ \varepsilon_{2i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{12} & \sigma_{22}^2 \end{pmatrix}\right)$$

The coefficient estimates of the treatment effect model are:

Variable	Estimates	Standard Errors	t-statistic
Intercept	0.9885	0.6809	1.4518
age	0.0054	0.0090	0.6042
educ	-0.0473	0.0448	-1.0559
black	-0.1709	0.2275	-0.7513
hisp	-0.5565	0.3101	-1.7948
married	0.1581	0.1676	0.9436
nodegree	-0.5407	0.1931	-2.7996
re74	-0.0222	0.0159	-1.3930
unem74	-0.3129	0.1862	-1.6807

For the outcome equation, the estimates are as follows:

Variable	Estimates	Standard Errors	t-statistic
Intercept	0.7586	2.5619	0.2961
age	0.0571	0.0450	1.2686
educ	0.4131	0.1789	2.3089
black	-2.2129	1.1777	-1.8790
hisp	0.0563	1.6482	0.0341
married	-0.1160	0.8819	-0.1315
re74	0.0823	0.0760	1.0823
re75	0.0517	0.1328	0.3895

The table for components of covariance matrix is as follows :

Parameter	Estimate	Standard Errors	t-statistic	p-value
ρ	0.0480	0.2826	0.170	0.8650
σ	6.4443	0.2226	28.949	0.00
σ^2	42.5296	—	—	—
σ_{12}	0.3096	1.8171	0.170	0.8650

⇒ The off-diagonal element σ_{12} (the covariance between the treatment and outcome disturbances) is small in magnitude and **statistically insignificant** ($t = 0.17$, $p = 0.865$) This indicates that there is no evidence of correlation between unobserved factors affecting program participation and those affecting earnings. Hence, selection on unobservables does not appear to be a concern in this model.

g). The coefficient estimate for γ the parameter for the variable train in the outcome equation is:

Variable	Estimate	SE	t
train	1.1940	2.9434	0.4057

The coefficient is not statistically significant and hence we cannot conclusively say that the job-training program is highly effective but based on the estimate it does have a slight positive effect on the wage. But overall due to the high SE and low t-statistic we **cannot** conclude that the job training program is effective.

Appendix

MATLAB Code

```

1 clear;
2 clc;
3 %% Question 2
4 % a)
5 data = readtable('Aptitude.xlsx');
6 head(data);
7 data.prog = categorical(data.prog);
8
9 % unique categories
10 categories(data.prog)
11
12
13 [counts, progNames] = histcounts(data.prog);
14
15
16 percentages = (counts / sum(counts)) * 100;
17
18 % results
19 disp('Program Type Summary:');
20 for i = 1:length(progNames)
21     fprintf('%s: %d observations (%.2f%%)\n', progNames{i}, counts(i),
22         percentages(i));
23 end
24 % indicator variable
25
26 data.academic = double(data.prog == 'academic');
27 data.general = double(data.prog == 'general');
28 data.vocational = double(data.prog == 'vocational');
29
30 % b) descriptive summary
31 vars = {'read', 'math', 'apt'};
32
33
34 summaryStats = table();
35
36 for i = 1:length(vars)
37     varName = vars{i};
38     x = data.(varName);
39
40
41     meanVal = mean(x, 'omitnan');
42     medianVal = median(x, 'omitnan');
43     stdVal = std(x, 'omitnan');
44     minVal = min(x);
45     maxVal = max(x);
46
47

```

```

48     summaryStats = [summaryStats;
49         table({varName}, meanVal, medianVal, stdVal, minVal, maxVal, ...
50             'VariableNames', {'Variable', 'Mean', 'Median', 'StdDev', 'Min'
51             , 'Max'})]];
52 end
53
54 disp('Descriptive Summary for read, math, and apt:');
55 disp(summaryStats);
56
57 % c) histogram of apt
58 programs = categories(data.prog);
59
60
61 figure;
62 hold on;
63
64
65 colors = lines(length(programs));
66
67
68 for i = 1:length(programs)
69     progType = programs{i};
70     subset = data.apt(data.prog == progType);
71
72     histogram(subset, 'FaceColor', colors(i,:), 'FaceAlpha', 0.5, ...
73         'DisplayName', progType, 'BinWidth', 20);
74 end
75
76
77 xlabel('Academic Aptitude (apt)');
78 ylabel('Frequency');
79 title('Histogram of Aptitude Scores by Program Type');
80 legend('Location', 'best');
81 grid on;
82 hold off;
83
84 % d) Correlation table and Scatter plot
85 X = data{:, {'read', 'math', 'apt'}};
86
87 % correlation matrix
88 corrMatrix = corr(X, 'Rows', 'complete');
89
90 % correlation table
91 varNames = {'read', 'math', 'apt'};
92 corrTable = array2table(corrMatrix, 'VariableNames', varNames, '
93     RowNames', varNames);
94 disp('Correlation Matrix:');
95 disp(corrTable);
96

```

```

97 figure;
98
99 subplot(1,3,1);
100 scatter(data.read, data.math, 'filled');
101 xlabel('Reading Score');
102 ylabel('Math Score');
103 title('Read vs Math');
104 grid on;
105
106 subplot(1,3,2);
107 scatter(data.read, data.apr, 'filled');
108 xlabel('Reading Score');
109 ylabel('Aptitude Score');
110 title('Read vs Apr');
111 grid on;
112
113 subplot(1,3,3);
114 scatter(data.math, data.apr, 'filled');
115 xlabel('Math Score');
116 ylabel('Aptitude Score');
117 title('Math vs Apr');
118 grid on;
119
120 sgtitle('Scatter Plots of Continuous Variables');
121
122 % e) Tobit Type-I Estimation (Right-Censored at 800)
123 % Dependent variable: apr
124 % Independent variables: read, math, program type (academic = base)
125 % Censoring: from above at 800 → apr = min(apr*, 800)
126
127 clear; clc;
128
129 data = readtable('Aptitude.xlsx');
130 y = data.apr;
131 read = data.read;
132 math = data.math;
133 prog = categorical(data.prog);
134
135 % Dummy variables for program type (academic = base)
136 D_general = double(prog == 'general');
137 D_vocational = double(prog == 'vocational');
138
139 % Design matrix
140 X = [ones(size(y)), read, math, D_general, D_vocational];
141 k = size(X, 2);
142
143 %% OLS initialization
144 ols_beta = X \ y; % OLS coefficient estimates
145 resid = y - X * ols_beta; % residuals
146 ols_sigma = std(resid); % residual standard deviation

```

```

147
148 % Initial parameter vector: [_OLS; log(_OLS)]
149 init_params = [ols_beta; log(ols_sigma)];
150
151 % Tobit log-likelihood setup
152 tobitLL = @(params) -sum(logL_tobit_right(params, y, X));
153
154 options = optimoptions('fminunc', ...
155     'Display', 'iter', 'Algorithm', 'quasi-newton', 'MaxIterations',
156     1000);
157 [params_hat, nll, exitflag, output, grad, hessian] = fminunc(tobitLL,
158     init_params, options);
159
160 beta_hat = params_hat(1:end-1);
161 sigma_hat = exp(params_hat(end));
162 se = sqrt(diag(inv(hessian)));
163 t_stats = params_hat ./ se;
164
165 varNames = {'Constant', 'Read', 'Math', 'General', 'Vocational', 'log(
166     sigma)'};
167 results = table(params_hat, se, t_stats, 'RowNames', varNames, ...
168     'VariableNames', {'Estimate', 'StdError', 'tStat'});
169
170 disp('-----');
171 disp('      Tobit Type I Model (Right-Censored at 800)');
172 disp('-----');
173 disp(results);
174 disp(['Sigma (std. dev of error term) = ', num2str(sigma_hat)]);
175 disp('-----');
176 disp(['Significance rule: |tStat| > 1.96 → significant at 5% level.']);
177 disp(['Negative log-likelihood = ', num2str(nll)]);
178 disp('-----');
179
180 % Custom Log-Likelihood Function (Right-Censoring)
181 function logL = logL_tobit_right(params, y, X)
182     beta = params(1:end-1);
183     sigma = exp(params(end)); % enforce sigma > 0
184     y_hat = X * beta;
185     c = 800; % upper-censoring threshold
186
187     % Identify censored observations
188     cens = (y >= c);
189     uncens = ~cens;
190
191     % Log-likelihood components
192     logL_cens = log(1 - normcdf((c - y_hat(cens)) / sigma)); % right
193     tail
194     logL_uncens = -0.5*log(2*pi) - log(sigma) - ...
195         0.5*((y(uncens) - y_hat(uncens)) / sigma).^2;

```

```

194
195     logL = zeros(size(y));
196     logL(cens) = logL_cens;
197     logL(uncens) = logL_uncens;
198 end
199
200 % f) Average Covariate Effect
201 % Compute Average Marginal Effects (AMEs)
202 c = 800; % upper censoring point
203 y_hat = X * beta_hat;
204 z = (c - y_hat) / sigma_hat;
205 Phi_z = normcdf(z);
206 phi_z = normpdf(z);
207
208 % Continuous covariates
209 AME_read = mean(Phi_z .* beta_hat(2)); % or mean(Phi_z)*beta_hat(2)
210 AME_math = mean(Phi_z .* beta_hat(3));
211
212 % Expected value function for upper-censored Tobit
213 E_y = @(XB) (XB .* normcdf((c - XB)/sigma_hat) ...
214             - sigma_hat * normpdf((c - XB)/sigma_hat) ...
215             + c * (1 - normcdf((c - XB)/sigma_hat)));
216
217 % Dummy variable (Vocational)
218 X_voc = X; X_voc(:,5) = 1;
219 X_acad = X; X_acad(:,5) = 0;
220
221 E_voc = E_y(X_voc * beta_hat);
222 E_acad = E_y(X_acad * beta_hat);
223 AME_voc = mean(E_voc - E_acad);
224
225 fprintf('Average Marginal Effect (Read): %.3f\n', AME_read);
226 fprintf('Average Marginal Effect (Math): %.3f\n', AME_math);
227 fprintf('Average Marginal Effect (Vocational): %.3f\n', AME_voc);
228
229 % (g) Diagnostic Plots: Residuals and Fitted Values (Simplified)
230
231 % Compute fitted latent and observed values
232 y_star_hat = X * beta_hat;
233 y_hat_obs = min(y_star_hat, 800); % censor at 800
234 residuals = y - y_hat_obs;
235
236 % Plot 1: Residuals vs Fitted Values
237 figure;
238 scatter(y_hat_obs, residuals, 40, 'b', 'filled'); hold on;
239 scatter(y_hat_obs(y >= 800), residuals(y >= 800), 40, 'r', 'filled');
240 xlabel('Fitted Values of apt (y-hat)', 'Interpreter', 'none');
241 ylabel('Residuals (y - y-hat)', 'Interpreter', 'none');
242
243 title('Residuals vs Fitted Values');
244 yline(0, '--k');

```

```

245 xline(800, '--r', 'Censoring Limit');
246 legend('Uncensored', 'Censored (apt=800)', 'Location', 'best');
247 grid on;
248
249 % Plot 2: Observed apt vs Fitted Values
250 figure;
251 scatter(y_hat_obs, y, 40, 'b', 'filled'); hold on;
252 scatter(y_hat_obs(y >= 800), y(y >= 800), 40, 'r', 'filled');
253 plot([0 800], [0 800], 'k--', 'LineWidth', 1);
254 xlabel('Fitted Values of apt (y-hat)', 'Interpreter', 'none');
255 ylabel('Observed Values of apt (y)', 'Interpreter', 'none');
256
257 title('Observed vs Fitted Values');
258 xline(800, '--r', 'Censoring Limit');
259 yline(800, '--r');
260 legend('Uncensored', 'Censored (apt=800)', '45° Line', 'Location', '
    best');
261 grid on;
262
263
264
265
266
267 %% Question 3
268
269 data = readtable('mroz.xlsx', 'Sheet', 'Data');
270
271 hours = data.hours;
272 educ = data.educ;
273 exper = data.exper;
274 hhours = data.hhours;
275 lfp = data.lfp;
276 kidsl6 = data.kidsl6;
277 kids618 = data.kids618;
278 mtr = data.mtr;
279 largecity = data.largecity;
280
281 n = height(data);
282
283 X = [ones(n,1), educ, exper, hhours];
284 Z = [ones(n,1), exper, kidsl6, kids618, mtr, largecity];
285
286 y = hours;
287 d = lfp;
288
289 kx = size(X,2);
290 kz = size(Z,2);
291
292 % OLS estimation
293 b = (X' * X) \ (X' * y); % OLS formula
294 yhat = X * b; % fitted values

```

```

295 res = y - yhat; % residuals
296
297 k = size(X,2);
298 sigma2 = (res' * res) / (n - k);
299 Varb = sigma2 * inv(X' * X);
300 se = sqrt(diag(Varb));
301 tstat = b ./ se;
302 pval = 2 * (1 - tcdf(abs(tstat), n - k));
303
304 fprintf('\n===== Q3(a): OLS Estimation =====\n');
305 varNames = {'Const', 'Educ', 'Exper', 'Hhours'};
306 for j = 1:k
307     fprintf('%-8s: %10.4f    SE=%8.4f    t=%8.3f    p=%8.4f\n', ...
308         varNames{j}, b(j), se(j), tstat(j), pval(j));
309 end
310 fprintf('-----\n');
311
312 %% Part b)
313
314 idx = hours > 0;
315 y_b = hours(idx);
316 X_b = [ones(sum(idx),1), educ(idx), exper(idx), hhours(idx)];
317 varNames = {'const', 'educ', 'exper', 'hhours'};
318
319 [n_b, k] = size(X_b);
320 beta = (X_b' * X_b) \ (X_b' * y_b);
321 u = y_b - X_b*beta;
322
323 sigma2 = (u' * u) / (n_b - k);
324 V = sigma2 * inv(X_b' * X_b);
325 se = sqrt(diag(V));
326 tstat = beta ./ se;
327
328 % --- Display results ---
329 disp('OLS on hours > 0');
330 fprintf('N (hours>0): %d\n\n', n_b);
331
332 T_out = table(varNames', beta, se, tstat, ...
333     'VariableNames', {'Variable', 'Coefficient', 'StdError', 'tStat'});
334 disp(T_out);
335
336 %% Part C)
337
338 X_probit = [ones(size(lfp)), exper, kidsl6, kids618, mtr, largecity];
339 k_probit = size(X_probit, 2);
340 llfun = @(b) - sum(lfp .* log(normcdf(X_probit*b)) + (1 - lfp) .* log(1
341     - normcdf(X_probit*b)));
342 opts = optimoptions('fminunc', 'Algorithm','quasi-newton', ...
343     'Display','iter', 'MaxFunctionEvaluations',1e5, 'MaxIterations',1e4
344     );

```

```

344 b0 = zeros(k_probit,1);
345 [bhat, fval, exitflag, output, grad, hess] = fminunc(llfun, b0, opts);
346
347 varb_probit = inv(hess);
348 se_probit = sqrt(diag(varb_probit));
349
350 tstat_probit = bhat ./ se_probit;
351
352 varNames = {'const','exper','kid16','kids618','MTR','largecity'};
353 disp(' ');
354 disp('Q3(c) PROBIT MODEL FOR LABOR FORCE PARTICIPATION');
355 T_out = table(varNames', bhat, se_probit, tstat_probit, ...
356     'VariableNames', {'Variable','Coefficient','StdError','tStat'});
357 disp(T_out);
358
359 %% Part D)
360
361 w_tilde = X_probit * bhat;
362 w_pdf = normpdf(w_tilde);
363 w_cdf = normcdf(w_tilde);
364
365 lambda = w_pdf ./ w_cdf;
366
367 lambda_mean = mean(lambda);
368 lambda_var = var(lambda);
369
370 fprintf('Sample mean of lambda = %.6f\n', lambda_mean);
371 fprintf('Sample var. of lambda = %.6f\n', lambda_var);
372
373 %% Part E)
374
375 X_ss = [ones(sum(idx), 1), educ(idx), exper(idx), hhours(idx), lambda(
    idx)];
376 [n_ss, k_ss] = size(X_ss);
377 y_ss = hours(idx);
378
379 beta_ss = (X_ss' * X_ss) \ (X_ss' * y_ss);
380 u_ss = y_ss - X_ss * beta_ss;
381
382 sigma2_ss = (u' * u) / (n - k);
383 var_ss = sigma2_ss * inv(X_ss' * X_ss);
384 se_ss = sqrt(diag(var_ss));
385 tstat_ss = beta_ss ./ se_ss;
386
387
388 varNames_ss = {'const','educ','exper','hhours','lambda'};
389 disp(' ');
390 disp('Q3(e): Heckman Two-Step Regression (Hours on and Covariates)');
391 T_out = table(varNames_ss', beta_ss, se_ss, tstat_ss, ...
392     'VariableNames', {'Variable','Coefficient','StdError','tStat'});
393 disp(T_out);

```

```

394
395 %% Part F
396
397 S = X_ss' * (diag(u_ss.^2)) * X_ss;
398 V_rob = inv(X_ss' * X_ss) * S * inv(X_ss' * X_ss);
399
400 beta_FGLS = (X_ss'*(diag(1./(u.^2 + eps)))*X_ss) \ (X_ss'*(diag(1./(u
    .^2 + eps)))*y_ss);
401
402 se_rob = sqrt(diag(V_rob));
403
404 t_fgls = beta_FGLS ./ se_rob;
405 % --- Display results for FGLS ---
406 varNames_fgls = {'const', 'educ', 'exper', 'hhours', 'lambda'};
407 T_out_fgls = table(varNames_fgls', beta_FGLS, se_rob, t_fgls, ...
408     'VariableNames', {'Variable', 'Coefficient', 'StdError', 'tStat'});
409 disp(' ');
410 disp('Q3(f): FGLS Regression Results');
411 disp(T_out_fgls);
412
413 %% Part g)
414
415 probit_fit = @(y, X) ...
416     fminunc(@(b) -sum( y.*log(max(normcdf(X*b), 1e-12)) + ...
417         (1-y).*log(max(1 - normcdf(X*b), 1e-12)) ), ...
418         zeros(size(X,2),1), ...
419         optimoptions('fminunc','Algorithm','quasi-newton','Display'
    , 'off', ...
420         'MaxFunctionEvaluations',1e5,'MaxIterations',1
    e4));
421
422
423 Xsel = X_probit;
424
425 B = 400;
426 k = numel(beta_ss);
427 boot_betas = NaN(k, B);
428
429 for b = 1:B
430     try
431         ii = randi(n, n, 1);
432
433         lfp_b = lfp(ii);
434         exper_b = exper(ii);
435         kids16_b = kids16(ii);
436         kids618_b = kids618(ii);
437         mtr_b = mtr(ii);
438         largecity_b = largecity(ii);
439         hours_b = hours(ii);
440         educ_b = educ(ii);
441         hhours_b = hhours(ii);

```

```

442
443     %Selection
444     Xsel_b = [ones(n,1), exper_b, kidsl6_b, kids618_b, mtr_b,
largacity_b];
445     b_sel_b = probit_fit(lfp_b, Xsel_b);
446     w_b = Xsel_b * b_sel_b;
447
448     lambda_b = normpdf(w_b) ./ max(normcdf(w_b), 1e-12);
449
450     idx_b = hours_b > 0;
451     if (sum(idx_b) <= k)
452         continue;
453     end
454     y_boot = hours_b(idx_b);
455     X_boot = [ones(sum(idx_b), 1), educ_b(idx_b), exper_b(idx_b),
hhours_b(idx_b), lambda_b(idx_b)];
456
457     beta_boot = (X_boot'*X_boot) \ (X_boot' * y_boot);
458     boot_betas(:, b) = beta_boot;
459     catch
460         continue
461     end
462 end
463
464 se_boot = std(boot_betas, 0, 2);
465
466 t_boot = beta_ss ./ se_boot;
467
468 disp('Q3(g): Heckman 2-step with Bootstrap Standard Errors (B = 400)');
469 fprintf('Valid bootstrap replications: %d (of %d)\n\n', size(boot_betas
,2), B);
470
471 T_out = table(varNames_ss', beta_ss, se_boot, t_boot, ...
472     'VariableNames', {'Variable','Coef','SE_boot','t_boot'});
473 disp(T_out);
474
475 %% Heckman model
476
477 sel1 = (lfp == 1);
478 sel0 = (lfp == 0);
479
480 beta0 = X(sel1,:) \ hours(sel1);
481 gamma0 = zeros(kz, 1);
482 lnsig0 = log(std(hours(sel1)));
483 arho0 = atanh(0.1);
484 theta0 = [beta0; gamma0; lnsig0; arho0];
485
486 function nll = heckman_nll(theta, X, Z, y, lfp, sel1, sel0)
487     kx = size(X,2);
488     kz = size(Z,2);
489     beta      = theta(1:kx);

```

```

490     gamma      = theta(kx+1:kx+kz);
491     ln_sigma    = theta(kx+kz+1);
492     atanh_rho   = theta(kx+kz+2);
493
494     sigma = exp(ln_sigma);
495     rho    = tanh(atanh_rho);
496     one_m_r2 = max(1 - rho^2, 1e-12);
497     srt = sqrt(one_m_r2);
498
499     XB  = X*beta;
500     ZG  = Z*gamma;
501
502
503     y1    = y(sel1);
504     XB1   = XB(sel1);
505     ZG1   = ZG(sel1);
506     a1    = (y1 - XB1) / sigma;
507
508     comp1 = -log(sigma) + log(normpdf(a1)) + log( max( normcdf( (ZG1 +
rho*a1)/srt ), 1e-12) );
509
510     ZG0  = ZG(sel0);
511     comp0 = log( max( normcdf(-ZG0), 1e-12) );
512
513     ll = sum(comp1) + sum(comp0);
514     nll = -ll;
515 end
516
517 obj = @(th) heckman_nll(th, X, Z, hours, lfp, sel1, sel0);
518
519 opts = optimoptions('fminunc', 'Algorithm','quasi-newton', 'Display', '
iter', ...
'MaxFunctionEvaluations',2e5, 'MaxIterations', 2e4);
521
522 [theta_hat, fval, exitflag, output, grad, hess] = fminunc(obj, theta0,
opts);
523
524 beta_he = theta_hat(1:kx);
525 gamma_he = theta_hat(kx+1:kx+kz);
526 lnsigma_hat = theta_hat(kx+kz+1);
527 atanhrho_hat = theta_hat(kx+kz+2);
528 sigma_hat = exp(lnsigma_hat);
529 rho_hat = tanh(atanhrho_hat);
530
531 V_he = inv(hess);
532 se_he = sqrt(diag(V_he));
533
534 pnames = [ ...
535     strcat("beta_", ["const","educ","exper","hhours"]), ...
536     strcat("gamma_", ["const","exper","kid16","kids618","mtr","
largecity"]), ...

```

```

537     "ln_sigma","atanh_rho" ]';
538
539 est = [beta_he; gamma_he; lnsigma_hat; atanhrho_hat];
540 tstats = est ./ se_he;
541
542 disp('Q3(h): Heckman Selection (FIML) Estimates');
543 T_out = table(pnames, est, se_he, tstats, 'VariableNames', {'Parameter'
    , 'Estimate', 'StdError', 'tStat'});
544 disp(T_out);
545
546
547 %%Question 4
548 clear; clc;
549
550 T = readtable('jtrain2.xlsx','Sheet','Data');
551
552 train      = T.train;
553 re78       = T.re78;
554 unem78     = T.unem78;
555 age        = T.age;
556 educ       = T.educ;
557 black      = T.black;
558 hisp       = T.hisp;
559 married    = T.married;
560 nodegree   = T.nodegree;
561 re74       = T.re74;
562 re75       = T.re75;
563 unem74     = T.unem74;
564
565 idx = all(~isnan([train,re78,unem78,age,educ,black,hisp,married,
    nodegree,re74,re75,unem74]),2);
566 train      = train(idx);
567 re78       = re78(idx);
568 unem78     = unem78(idx);
569 age        = age(idx);
570 educ       = educ(idx);
571 black      = black(idx);
572 hisp       = hisp(idx);
573 married    = married(idx);
574 nodegree   = nodegree(idx);
575 re74       = re74(idx);
576 re75       = re75(idx);
577 unem74     = unem74(idx);
578
579 n = length(train);
580
581 fprintf('(a) Fraction trained: %.4f (n=%d)\n', mean(train), n);
582 m1 = mean(re78(train==1)); m0 = mean(re78(train==0));
583 fprintf('(b) Mean re78 | train=1: %.4f\n      Mean re78 | train=0: %.4f\n
    Diff: %.4f\n', m1, m0, m1-m0);
584 u1 = mean(unem78(train==1)); u0 = mean(unem78(train==0));

```

```

585 fprintf('(c) Unemp78 | train=1: %.4f\n      Unemp78 | train=0: %.4f\n',
        u1, u0);
586
587 X1 = [ones(n,1), age, educ, black, hisp, married, nodegree, re74,
        unem74];
588 p1 = size(X1,2);
589 X2_base = [ones(n,1), age, educ, black, hisp, married, re74, re75];
590 p2 = size(X2_base,2);
591 y = re78; d = train;
592
593 g0 = zeros(p1,1);
594 pbar = min(max(mean(d),1e-3), 1-1e-3);
595 g0(1) = log(pbar/(1-pbar));
596
597 Bols    = [X2_base, d] \ y;
598 beta0   = Bols(1:p2);
599 alpha0  = Bols(end);
600 resid0  = y - [X2_base, d]*Bols;
601 logsig0 = log(std(resid0));
602 atanh_rho0 = atanh(0.0);
603
604 theta0 = [g0; beta0; alpha0; logsig0; atanh_rho0];
605
606 loglik = @(theta) -sum( perObsLL(theta, X1, X2_base, y, d) );
607
608 opts = optimoptions('fminunc','Algorithm','quasi-newton', ...
609     'Display','iter','MaxFunctionEvaluations',1e5,'MaxIterations',1e4,
610     ...
611     'OptimalityTolerance',1e-6,'StepTolerance',1e-8);
612 [thetahat, fval, exitflag, output, grad, H] = fminunc(loglik, theta0,
        opts);
613
614 gamma    = thetahat(1:p1);
615 beta     = thetahat(p1+(1:p2));
616 alpha    = thetahat(p1+p2+1);
617 logsig   = thetahat(p1+p2+2);
618 atanhhr  = thetahat(p1+p2+3);
619 sigma    = exp(logsig);
620 rho      = tanh(atanhr);
621
622 V = inv(H);
623 se  = sqrt(diag(V));
624 se_g = se(1:p1);
625 se_b = se(p1+(1:p2));
626 se_a = se(p1+p2+1);
627 se_logsig = se(p1+p2+2);
628 se_atanhhr = se(p1+p2+3);
629
630 se_sigma = sigma * se_logsig;
631 se_rho   = (1 - rho^2) * se_atanhhr;

```

```

632
633 pnorm = @ (z) 0.5*(1+erf(z./sqrt(2)));
634 pval = @ (z) 2*(1 - pnorm(abs(z)));
635
636 z_g = gamma ./ se_g;
637 z_b = beta ./ se_b;
638 z_a = alpha ./ se_a;
639 z_sig = sigma ./ se_sigma;
640 z_rho = rho ./ se_rho;
641
642 p_g = pval(z_g);
643 p_b = pval(z_b);
644 p_a = pval(z_a);
645 p_sig = pval(z_sig);
646 p_rho = pval(z_rho);
647
648 fprintf('\n===== (e)-(f) MLE results with p-values
        =====\n');
649 fprintf('alpha (train): %.4f SE: %.4f z: %.4f p: %.4f\n', alpha,
        se_a, z_a, p_a);
650 fprintf('sigma: %.4f (SE %.4f) z: %.4f p: %.4f\n', sigma, se_sigma,
        z_sig, p_sig);
651 fprintf('rho: %.4f (SE %.4f) z: %.4f p: %.4f\n\n', rho, se_rho,
        z_rho, p_rho);
652
653 labels_sel = {'Intercept','age','educ','black','hisp','married','
        nodegree','re74','unem74'};
654 sel_tab = [gamma, se_g, z_g, p_g];
655 disp('Treatment (probit) coefficients: [Estimate SE z
        p]');
656 disp([labels_sel', num2cell(round(sel_tab,4))]);
657
658 labels_out = {'Intercept','age','educ','black','hisp','married','re74',
        're75','train'};
659 coef_out = [beta; alpha];
660 se_out = [se_b; se_a];
661 z_out = coef_out ./ se_out;
662 p_out = pval(z_out);
663
664 out_tab = [coef_out, se_out, z_out, p_out];
665 disp('Outcome equation (re78), INCLUDING train: [Estimate SE
        z p]');
666 disp([labels_out', num2cell(round(out_tab,4))]);
667
668 function li = perObsLL(theta, X1, X2b, y, d)
669     p1 = size(X1,2);
670     p2 = size(X2b,2);
671     gamma = theta(1:p1);
672     beta = theta(p1+(1:p2));
673     alpha = theta(p1+p2+1);
674     logsig = theta(p1+p2+2);

```

```

675     atanh_r = theta(p1+p2+3);
676     sigma    = exp(logsig);
677     rho      = tanh(atanh_r);
678     mu       = X2b*beta + alpha.*d;
679     eps      = (y - mu) ./ sigma;
680     w_i      = X1*gamma;
681     c        = sqrt(max(1e-12, 1 - rho^2));
682     arg      = (w_i + rho.*eps) ./ c;
683     logfy    = -0.5*log(2*pi) - logsig - 0.5*(eps.^2);
684     Phi1     = 0.5*(1+erf(arg./sqrt(2)));
685     Phi1     = max(min(Phi1,1-1e-12),1e-12);
686     Phi0     = 1 - Phi1;
687     Phi0     = max(min(Phi0,1-1e-12),1e-12);
688     li       = logfy + d.*log(Phi1) + (1-d).*log(Phi0);
689 end
690
691 sigma_12 = rho * sigma;
692 se_sigma12 = sqrt( (sigma^2)*(se_rho^2) + (rho^2)*(se_sigma^2) );
693 fprintf('sigma_12 (covariance term) = %.4f  ( SE %.4f)\n', sigma_12,
        se_sigma12);

```